Lecture 3

In Class Exercises

Exercise 1

- Indicate, for each pair of expressions (A, B) in the table below whether A is Θ, Ω or O of B. Assume that $k \geq 1$, $\varepsilon > 0$, and $c > 1$ are constants. Write your answer in the form of the table with "yes" or "no" written in each box.

	A	B	O	Ω	Θ
a.	$lg^k n$	n^{ϵ}			
b.	n^k	c^n			
c.	$\sqrt{n}$	$n^{\sin n}$			
d.	2^n	$2^{n/2}$			
e.	$n^{\lg c}$	$c^{\lg n}$			
f.	$\lg(n!)$	$\lg(n^n)$			

Exercise 2

Is $2^{n+1} = O(2^n)$? Is $2^{2n} = O(2^n)$?

- Supplement your answers by providing a proof according to the O theorem (e.g. find a constant c…).

Exercise 3

- Prove the following theorem:

Theorem 3.1

For any two functions $f(n)$ and $g(n)$, we have $f(n) = \Theta(g(n))$ if and only if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$. ■

Proof by induction (Recap)

- There are lg n + 1 levels to a recursion tree for an input of size n.
- Base case: $n = 1 \Rightarrow 1$ level, and $\lg 1 + 1 = 0 + 1 = 1$.
- Inductive hypothesis is that a tree for a problem size of 2^i has
  $\lg 2^i + 1 = i + 1$ levels.
- Because we assume that the problem size is a power of 2, the next problem size up after 2^i is 2^{i+1}
- A tree for a problem size of 2^{i+1} has one more level than the size-2^i tree => $i + 2$ levels.
- Since $\lg 2^{i+1} + 1 = i + 2$, we are done with the inductive argument.

Exercise 4

- Use mathematical induction to show that when n ≥ 2 is an exact power of 2, the solution of the recurrance

$$T(n) = \begin{cases} 2 & \text{if } n = 2\,, \\ 2T(n/2) + n & \text{if } n > 2 \end{cases}$$

- is $T(n) = nlgn.$